Reverse isoperimetric problem under curvature constraints

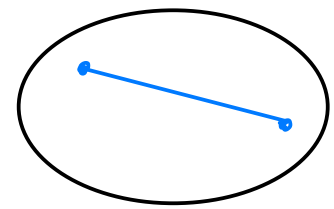
(joint with K. Drach)

$K \subseteq \mathbb{R}^n$ convex body = compact convex set with non-empty interior in $\mathbb{R}^n$

$|\partial K|$ – fixed

$|K|$ – max

Euclidean ball



Reverse question:

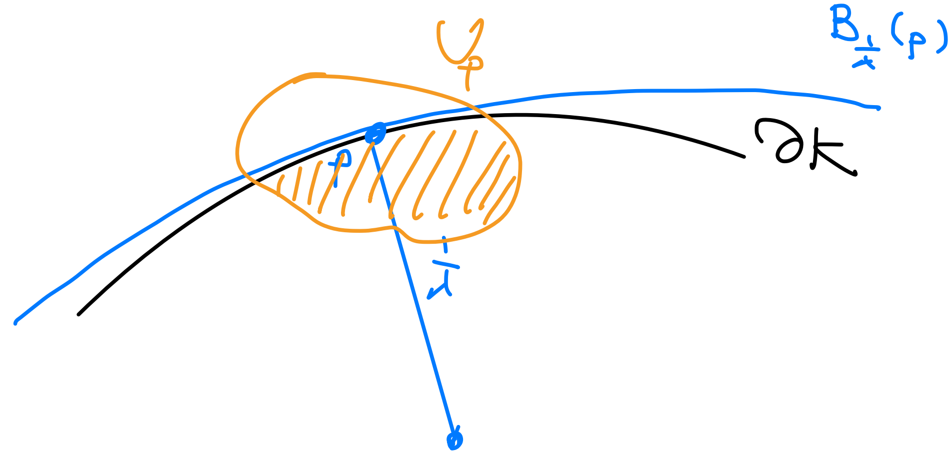
$|\partial K|$ - fixed

$|K|$ - min

flat pancakes

__Def:__ $K \subset \mathbb{R}^n$ convex body, $\lambda > 0$.

K is λ-convex if



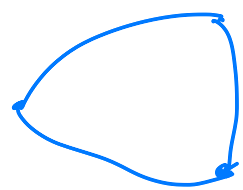
$$K \cancel{\cap U_p} \subseteq B_{\frac{1}{\lambda}}(p)$$

Blaschke's rolling thm guarantees that this local condition is in fact global

$\iff$ principal curvatures $k_i \geq \lambda$.

Example: $\underbrace{\text{finite intersection of ball of radius } \tfrac{1}{\lambda}}_{\text{ball polytopes}}$.

lens



__Thm__: $K \subset \mathbb{R}^3$ a λ-convex body in $\mathbb{R}^3$

L = lens = intersection of two balls of radius $\frac{1}{\lambda}$

If $|\partial K| = |\partial L|$ then $|K| \geq |L|$.

"=" $K = L$

__Conjecture__ (Borisenko, 2010; Bezdek, 2015): thm is true in $\mathbb{R}^n$.

$n=2$ Borisenko - Drach, 2014

Fodor - Kurusa - Vígh, 2016

$n \geq 4$ open

Sketch of proof:

ball-polytopes

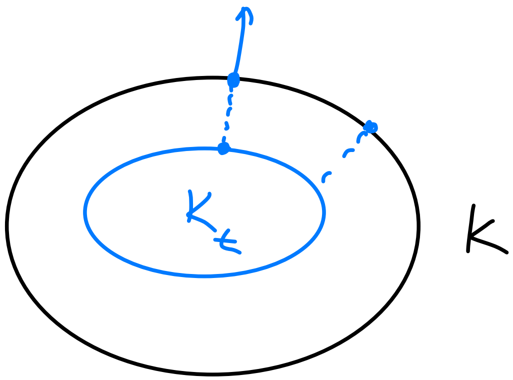
$\lambda = 1$

vector addition

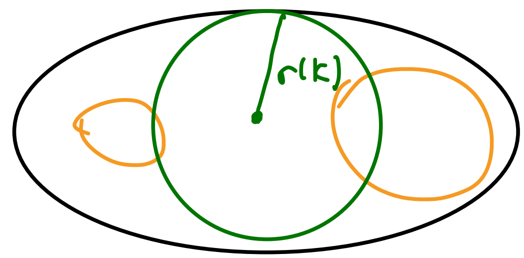
Def: $K_t = \{ x \in \mathbb{R}^n : x + tB \subseteq K \}$

unit ball

inner parallel body



$0 \le t \le \underbrace{r(K)}_{\text{inradius of } K}$



Matheron '78:

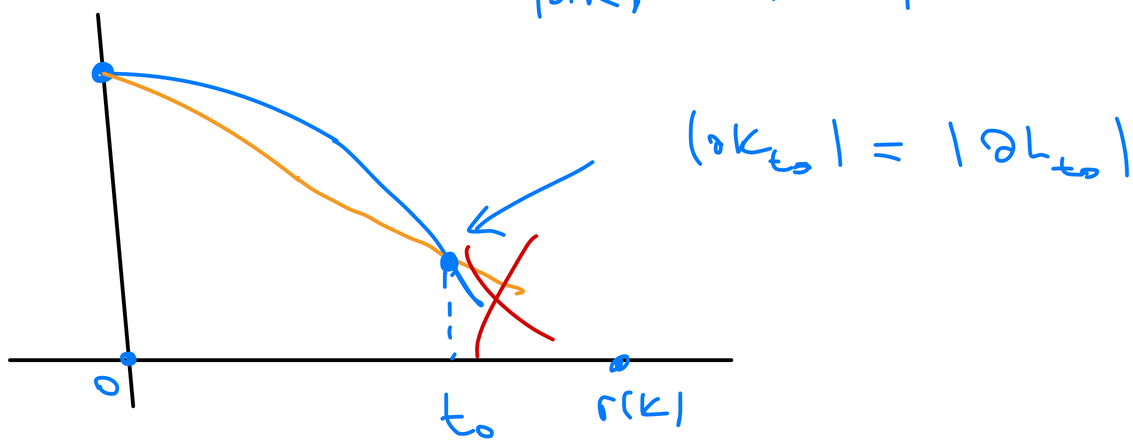
$$|K| = \int_0^{r(K)} |\partial K_t| \, dt.$$

$$|K| = \int |\partial K_t| \ge \int |\partial L_t| = |L|$$

wts $|\partial K_t| \geq |\partial L_t| \quad \forall\, t \in (0, r(K))$

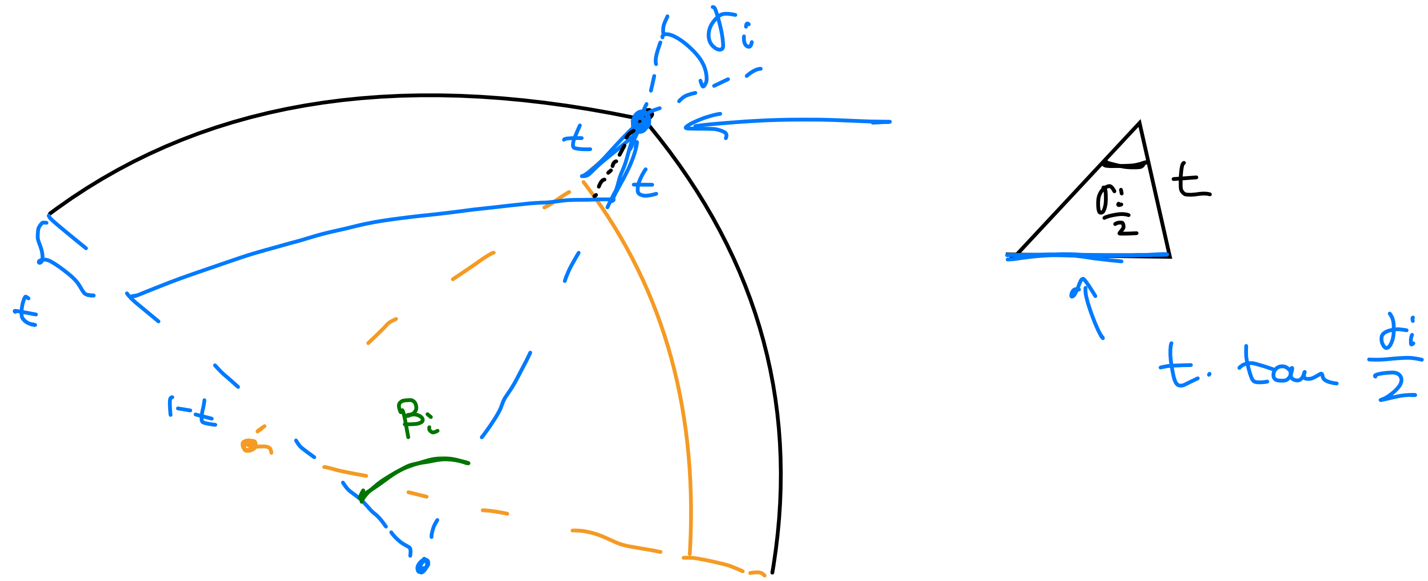
Claim: $|\partial K_t| > |\partial L_t|$ for small t around 0.

$|\partial K| = |\partial L|$



wts $\left.\dfrac{d}{dt} |\partial K_t|\right|_{t=0} > \left.\dfrac{d}{dt} |\partial L_t|\right|_{t=0^+}$

$\mathbb{R}^2$:



$$|\partial K_t| = \sum (1-t)\beta_i - 2t \sum \tan \frac{\gamma_i}{2} + O(t^2)$$

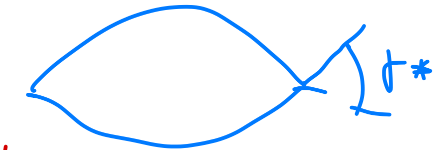
Note $\quad |\partial K| = \sum \beta_i$

$$\frac{d}{dt} |\partial K_t| \Big|_{t=0^+} = - \sum \beta_i - 2 \sum \tan \frac{\gamma_i}{2} .$$

$$\|$$

$$|\partial K| = |\partial L|$$

$$\frac{d}{dt}\left|\partial L_t\right|\Big|_{t=0^+} = -|\partial L| - 4\tan\frac{\gamma_*}{2}$$



maximize $$\sum \tan\frac{\gamma_i}{2} \le 2\tan\frac{\gamma_*}{2}$$

(I)

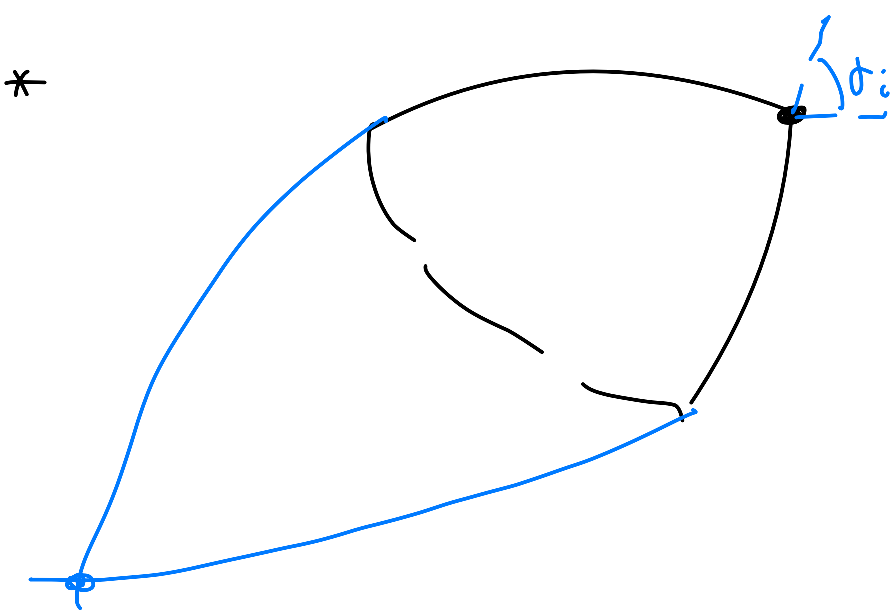
$$2\pi = |\partial K| + \sum \gamma_i$$

$$2\pi = |\partial L| + 2\gamma_*$$

$$\Rightarrow \sum \gamma_i = 2\gamma_* .$$

(II)

$$\max \gamma_i \le \gamma_*$$



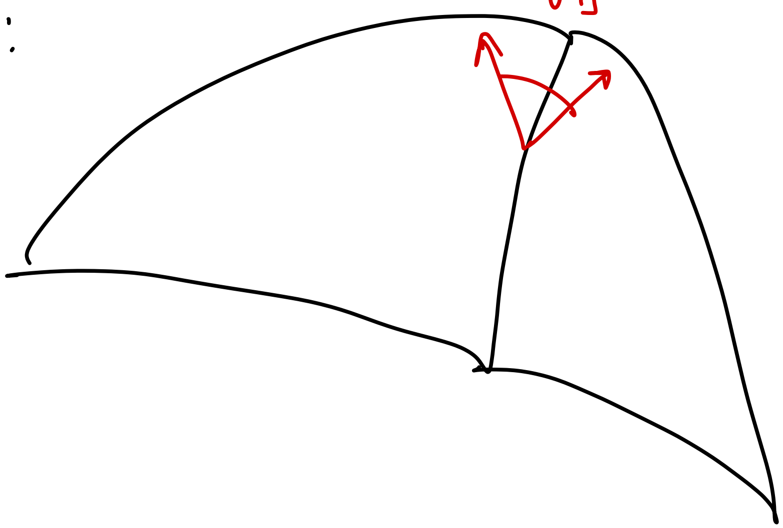
$$\sum \tan \frac{\gamma_i}{2} = \sum \frac{\tan \frac{\gamma_i}{2}}{\frac{\gamma_i}{2}} \cdot \frac{\gamma_i}{2}$$

increasing

$$\overset{\text{II}}{\leq} \sum \frac{\tan \frac{\gamma_*}{2}}{\frac{\gamma_*}{2}} \cdot \frac{\gamma_i}{2} = \gamma_* \cdot \frac{\tan \frac{\gamma_*}{2}}{\frac{\gamma_*}{2}}$$

$$= 2 \tan \frac{\gamma_*}{2}$$

$\mathbb{R}^3$:



β_i = area of each facet

γ_{ij} = angle between facets

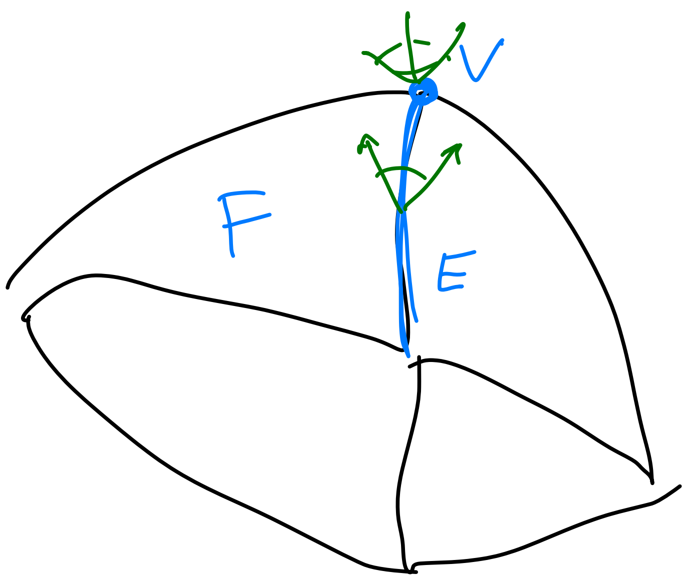
l_{ij} = length of edge

$$\frac{d}{dt} \left| \partial K_t \right| \Big|_{t=0^+} = -2 \, |\partial K| - 2 \sum l_{ij} \tan \frac{\theta_{ij}}{2}$$

$$\underset{|\partial L|}{\overset{\Downarrow}{|\partial K|}}$$

maximize $\sum l_{ij} \tan \frac{\theta_{ij}}{2}$

Gauss – Bonnet thm $\displaystyle\int_{\partial K} \kappa \, d\sigma_k = 4\pi$



$$\overbrace{\sum \beta_i}^{=|\partial K|} + 2\sum l_{ij} \tan\frac{\alpha_{ij}}{2} + \underbrace{V}_{\geq 0} = 4\pi$$

$$2\sum l_{ij} \tan\frac{\alpha_{ij}}{2} = 4\pi - |\partial K| - \underbrace{V}_{= 0 \text{ if no vertices}}$$

the only finite intersection of balls without vertices is lens